CONSTANT PRESSURE BLOCKING FILTRATION LAWS—APPLICATION TO POWER-LAW NON-NEWTONIAN FLUIDS

By J. HERMIA

Institut de Genie Chimique UCL, Louvain-la-Neuve, Belgium

A physical model is presented to derive the so-called 'intermediate blocking' law which has so far been considered totally empirical. The mechanism proposed is still based on pore sealing, just as is assumed for 'complete blocking' filtration, involving geometric considerations as to particles and filter membrane interaction. In addition to this, the probability of pore blocking has been considered in this new model. The 'standard blocking' and 'filter cake' laws are recalled and compared with the first two blocking filtration models. Application to power-law non-Newtonian fluids is made for constant pressure filtration.

INTRODUCTION

Blocking filtration laws were first studied by Hermans and Bredée[1]. Gonsalves[2] made a critical study of the physical models used to derive these laws. Grace[3] made a thorough analysis of the blocking laws in relation to the performance of the filter media to be used, with particular reference to the standard blocking law. In an earlier paper by the author[4], a physical model was presented to derive the so-called 'intermediate blocking' law which had previously been considered totally empirical.

In a more recent paper, Shirato *et al*[5] extended the study of the blocking laws to the case of power-law non-Newtonian fluids, excepting the case of the intermediate law. In this paper, a derivation of all blocking laws is presented, including the intermediate, and the models are applied to power-law non-Newtonian fluids for constant pressure filtration.

COMPLETE BLOCKING FILTRATION LAW

Let us consider a filter medium such as those encountered in liquid clarification: the porous material (called the membrane) can be a sheet of paper, a thin layer of filter-aid, a piece of cellulose, asbestos, cardboard or suchlike. We assume that each particle reaching the membrane participates in the blocking phenomenon by pore sealing, which leads to the assumption that particles are not superimposed one upon the other.

With such an assumption, we can state that when a filtrate volume V has been filtered, it has blocked a portion of the membrane surface area equal to σV. To evaluate σ, let d be the equivalent particle diameter (diameter of the sphere of the same volume). The volume of a particle is then $\pi d^3/6$ and its total surface area is $\pi d^2/\psi$ where $0<\psi\leqslant 1$ is a form factor depending on particle shape. This surface projected on the membrane is approximately equal to $\pi d^2/4\psi$.

The volume of the particles held in the slurry volume V^* is equal to $V^*\gamma_s s/\gamma_0$ where γ_s is the slurry density, γ_0 is the solid particles density and s is the mass fraction of solids in the slurry.

Then, we can find the surface area of the total amount of particles, projected onto the filter membrane:

$$\left(\frac{V^*\gamma_s s}{\gamma_0}\right)\left(\frac{6}{\pi d^3}\right)\left(\frac{\pi d^2}{4\psi}\right) = \sigma V^* \simeq \sigma V$$

Indeed, for dilute suspensions such as those encountered in clarification problems, s is small ($\leqslant 0.1\%$) and therefore the slurry volume V^* and the filtrate volume V are almost the same. From the above expression, it follows that

$$\sigma = 1.5\frac{\gamma_s s}{\gamma_0 d\psi} \quad (\mathrm{m}^{-1})$$

and is a characteristic of the slurry.

Darcy's law states that $Q = PA/\mu R$, so the initial flow rate will be $Q_0 = PA_0/\mu R$, while at time t the flow rate will be given by

$$Q = \frac{PA_t}{\mu R} \tag{1}$$

where $A_t = A_0 - \sigma V$, assuming a constant pressure filtration.

We then obtain $Q = Q_0 - P\sigma V/\mu R$ which can be put in the following form:

$$K_b V = Q_0 - Q \tag{2}$$

which is a linear relationship between V and Q, the constant K_b being given by

$$K_b = \frac{P\sigma}{\mu R} = \frac{Q_0\sigma}{A_0} = u_0\sigma\ (\mathrm{s}^{-1}) \tag{3}$$

Integrating (2) with respect to time yields

$$Q = \frac{\mathrm{d}V}{\mathrm{d}t} = Q_0 - K_b V$$

hence

$$Q = Q_0 \mathrm{e}^{-K_b t} \tag{4}$$

which is the $Q = f(t)$ relationship, while

$$K_b V = Q_0(1-\mathrm{e}^{-K_b t}) \tag{5}$$

is the $V = f(t)$ relationship.

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For a constant pressure filtration, we shall define the 'resistance coefficient' as the rate of variation with respect to filtrate volume of the instantaneous resistance to filtration which can be measured by the inverse of the flow rate:

$$\text{Resistance coefficient} = \frac{d}{dV}\left(\frac{1}{Q}\right) = \frac{d^2 t}{dV^2}$$

Introducing (2) in this definition, we can find:

$$\frac{d^2 t}{dV^2} = \frac{d}{dV}\left(\frac{1}{Q_0 - K_b V}\right) = \frac{K_b}{Q^2}$$

hence

$$\frac{d^2 t}{dV^2} = K_b\left(\frac{dt}{dV}\right)^2 \tag{6}$$

which can be considered as the characteristic form of the complete blocking filtration law under constant pressure.

INTERMEDIATE BLOCKING FILTRATION LAW

As in the previous section, we still consider that any time a solid particle reaches an open pore, it seals it. Nevertheless, in this case we adopt a less restrictive model in such a way that we allow particles to settle on other particles. In other words, we do not state that each particle does necessarily block a pore but on the contrary we evaluate the *probability* for a particle to block a pore.

Let A_0 be the initial active filter membrane surface area Let V^* be a small prefilter volume containing particles whose projected area on the membrane is given by $\Sigma = \sigma V^* \simeq \sigma V$ (see previous section). Let us assume Δt is the time required to filter the first batch V^*.

After this slurry volume has passed through the membrane, the active surface is reduced to $(A_0 - \Sigma)$. For a next batch, the volume of filtrate will be reduced to $V(A_0 - \Sigma)/A_0$ and will allow the deposit of solid particles whose surface area projected on the membrane will be $\Sigma(A_0 - \Sigma)/A_0$.

If we assume that the suspension is perfectly homogeneous, the second particle layer has an equal probability to settle on the first layer as on the surface which was left free. The *increment* of blocked area will then be proportional to the surface area which was unblocked and therefore will be given by

$$\Sigma\left(\frac{A_0 - \Sigma}{A_0}\right)\left(\frac{A_0 - \Sigma}{A_0}\right)$$

At the end of this second step, the free surface is equal to

$$(A_0 - \Sigma) - \Sigma\left(\frac{A_0 - \Sigma}{A_0}\right)^2$$

For recurrent successive batches (at time $t - \Delta t$, t, $t + \Delta t, \ldots$), we can state that the free unblocked surface area at time t, *i.e.* A_t, is equal to that at time $t - \Delta t$, i.e. $A_{t-\Delta t}$, from which we deduce the new blocked area which is proportional to both the free surface at time $(t - \Delta t)$ and to the filtrate volume passing through this free surface.

Let us now allow time lags to tend towards a limit dt:

$$A_{t+dt} = A_t - \sigma(Q_t \, . \, dt)\left(\frac{A_t}{A_0}\right)$$

Introducing $Q = PA/\mu R$ in this relation, we obtain

$$dA = -\frac{P\sigma}{\mu R A_0} A^2 \, dt$$

Integration of this last relation yields

$$A = \frac{A_0}{1 + (\sigma P/\mu R)\, t} \tag{7}$$

which implies

$$Q = \frac{Q_0}{1 + (\sigma P/\mu R)\, t}$$

which can be put in the following form:

$$K_i t = \frac{1}{Q} - \frac{1}{Q_0} \tag{8}$$

where K_i (m^{-3}) is equal to

$$\frac{\sigma P}{\mu R Q_0} = \frac{\sigma}{A_0} \tag{9}$$

It is then easy to deduce from (8) that

$$Q = Q_0 e^{-K_i V} \tag{10}$$

which is the $Q = f(V)$ relationship and

$$K_i V = \ln(1 + K_i Q_0 t) \tag{11}$$

which is the $V = f(t)$ relationship.

The resistance coefficient will be given by

$$\frac{d^2 t}{dV^2} = \frac{d}{dV}\left(\frac{1}{Q}\right) = \frac{d}{dV}\left(\frac{1}{Q_0} e^{K_i V}\right) = \frac{K_i}{Q}$$

hence

$$\frac{d^2 t}{dV^2} = K_i \frac{dt}{dV} \tag{12}$$

which can be considered as the characteristic form of the intermediate blocking filtration law for constant pressure.

STANDARD BLOCKING FILTRATION LAW

In deriving the standard blocking law, it is assumed that pore volume decreases proportionally to filtrate volume by particle deposit on the pore walls. The filter membrane is assumed to consist of a set of pores of constant diameter. The length of the pores being also assumed constant, the decrease of pore volume will be equal to the decrease of pore section.

A mass balance on solid particles yields

$$N^*(-2\pi r \, dr) L = C \, dV$$

where N^* is the number of pores, L the pore length and C the volume of particles deposited by unit volume of filtrate.

Integrating this relation we obtain

$$N^* \pi (r_0{}^2 - r^2) L = CV$$

Making use of Poiseuille's equation to find the initial flow-rate:

$$Q_0 = N^*\left(\frac{\pi}{8}\frac{r_0^4 P}{\mu L}\right) \tag{13}$$

it is possible to derive the following relation:

$$Q = Q_0\left(1-\frac{K_s V}{2}\right)^2 \tag{14}$$

where the constant K_s (m^{-3}) is equal to

$$\frac{2C}{\pi L N^* r_0^2} = \frac{2C}{LA_0} \tag{15}$$

Integrating (14) yields the $V=f(t)$ relationship:

$$\frac{K_s t}{2} = \frac{t}{V} - \frac{1}{Q_0} \tag{16}$$

It is also possible to derive the $Q=f(t)$ relationship:

$$Q = \frac{Q_0}{(1+\frac{1}{2}K_s Q_0 t)^2} \tag{17}$$

The resistance coefficient will be given by

$$\frac{d^2 t}{dV^2} = \frac{d}{dV}\left[\frac{1}{Q_0(1-\frac{1}{2}K_s V)^2}\right] = K_s Q_0^{1/2}\left(\frac{1}{Q}\right)^{3/2}$$

The characteristic form of the standard blocking filtration law for constant pressure can thus be written in the following form:

$$\frac{d^2 t}{dV^2} = K_s'\left(\frac{dt}{dV}\right)^{3/2} \tag{18}$$

where $K_s' = K_s Q_0^{1/2}$.

CAKE FILTRATION LAW

Let $Q_0 = PA/\mu R_0$ be the initial flow rate and $Q = PA/\mu R_t$ the flow rate at time t. The filter resistance is composed of the membrane resistance R_0 and the cake resistance: $R_t = R_0 + \alpha W/A$.

A mass balance on the cake yields $W = V\gamma s/(1-ms)$, so we can write

$$R_t = R_0 + \frac{\alpha\gamma s V}{(1-ms)A} = R_0\left[1+\frac{\alpha\gamma s V}{AR_0(1-ms)}\right]$$

and the following relation can be derived:

$$R_t = R_0(1+K_c Q_0 V)$$

where

$$K_c = \frac{\alpha\gamma s}{AR_0 Q_0(1-ms)} = \frac{\alpha\gamma s\mu}{A^2 P(1-ms)} \quad (s\,m^{-6}) \tag{19}$$

It is easy to verify that $K_c = 2/K_R$ where K_R is the classical Ruth's constant

$$(V+V_c)^2 = K_R(t+t_c) = \frac{2A^2 P(1-ms)}{\mu\gamma s\alpha}(t+t_c)$$

We can then derive

$$Q_t = \frac{PA}{\mu R_t} = \frac{PA}{\mu R_0(1+K_c Q_0 V)} = \frac{Q_0}{1+K_c Q_0 V}$$

which yields

$$K_c V = \frac{1}{Q} - \frac{1}{Q_0} \tag{20}$$

Integrating (20) gives

$$\frac{K_c V}{2} = \frac{t}{V} - \frac{1}{Q_0} \tag{21}$$

which is the $V=f(t)$ relationship, while the $Q=f(t)$ can be written:

$$Q = \frac{Q_0}{(1+2K_c Q_0^2 t)^{1/2}} \tag{22}$$

The resistance coefficient is given by

$$\frac{d^2 t}{dV^2} = \frac{d}{dV}\left(\frac{1+K_c Q_0 V}{Q_0}\right) = K_c$$

hence

$$\frac{d^2 t}{dV^2} = K_c \tag{23}$$

can be considered as the characteristic form of the cake filtration law for constant pressure.

APPLICATION TO POWER-LAW NON-NEWTONIAN FLUIDS

We have seen that the characteristic form of the blocking filtration laws is

$$\frac{d^2 t}{dV^2} = k\left(\frac{dt}{dV}\right)^n$$

for constant pressure filtration.

It may be seen in the Table that for all types of filtration laws, except for the intermediate, the constant k always depends on the initial flow rate (Q_0 or u_0). This is the reason why, when a non-Newtonian fluid is dealt with, constants k and n are related to the parameters which define the properties of this non-Newtonian fluid, as shown by Shirato *et al*[5].

Table

Law	k	n	See equation
Cake filtration	$K_c = \frac{\alpha\gamma s}{AR_0 Q_0(1-ms)}$	0	(23)
Intermediate	$K_i = \frac{\sigma}{A_0}$	1	(12)
Standard law	$K_s' = \frac{2C}{LA_0}Q_0^{1/2}$	$\frac{3}{2}$	(18)
Complete blocking	$K_b = u_0\sigma$	2	(6)

Indeed, let us consider the Rabinowitsch–Mooney equation applied to the flow in a cylindrical pore and written as

$$\frac{u}{r} = \frac{1}{\tau_W^3}\int_0^{\tau_W} \tau^2 f(\tau)\,d\tau \tag{24}$$

which relates the velocity at radius r to the shear stress τ at the radius and to the shear at the wall τ_W. This equation is valid for the flow of Newtonian and time-independent non-Newtonian fluids.

In the case of power-law non-Newtonian fluids, the rheological equation is given by

$$f(\tau) = \left(\frac{\tau}{K}\right)^{1/N} \tag{25}$$

Substituting (25) in (24) and integrating yields the relation between the velocity and the shear stress at the wall:

$$\frac{u}{r} = \frac{1}{\tau_W{}^3}\int_0^{\tau_W} \tau^2\left(\frac{\tau}{K}\right)^{1/N} d\tau = K^N \frac{N}{3N+1}\tau_W^{1/N} \tag{26}$$

Now the shear stress at the wall can be related to the pressure drop according to

$$(2\pi rL)\tau_W = P\pi r^2$$

hence

$$\tau_W = Pr/2L \tag{27}$$

Substituting this value of τ_W in (26) yields

$$u = \frac{N}{3N+1}\left(\frac{P}{2LK}\right)^{1/N} r^{(N+1)/N} \tag{28}$$

If there is no change in pore diameter, $r = r_0$. For the filtration of a given fluid (N and K fixed), under a constant pressure ($P = P_0$) on a given membrane (L and r fixed), the velocity in a pore is exactly determined by equation (28) and the total flow rate will be determined by the number of active pores which remain open in the membrane. This will be the case for complete and intermediate blocking filtration laws; on the other hand, for the standard law, the pore radius r diminishes as filtration proceeds while in cake filtration, cake thickness L increases.

Cake Filtration

In the case of cake filtration, equation (28) shows that for a constant pressure, u is proportional to $L^{-1/N}$, which means that Q is proportional to $V^{-1/N}$. Equation (20) is then transformed into

$$K_c V = \left(\frac{1}{Q}\right)^N - \left(\frac{1}{Q_0}\right)^N$$

The characteristic equation will then take the following form:

$$\frac{d^2 t}{dV^2} = \frac{d}{dV}\left(\frac{1}{Q}\right) = \frac{d}{dV}(K_c V)^{1/N} = \frac{K_c}{N}\left(\frac{1}{Q}\right)^{1-N}$$

hence

$$\frac{d^2 t}{dV^2} = \frac{K_c}{N}\left(\frac{dt}{dV}\right)^{1-N} \tag{29}$$

which is the new characteristic form for a constant pressure cake filtration of a power-law non-Newtonian fluid. We see that both the clogging constant and the exponent of the characteristic form depend on N.

Standard Blocking Law

Equation (28) shows that u is proportional to $r^{(N+1)/N}$ and thus the flow-rate for a constant pressure filtration will be proportional to $N^*(\pi r^2) r^{(N+1)/N}$. We shall then write

$$Q = \frac{N}{3N+1}\left(\frac{P_0}{2KL}\right)^{1/N} N^* \pi r^2 r^{(N+1)/N}$$

which can be put in the following form:

$$Q = \beta r^{(3N+1)/N} \tag{30}$$

where

$$\beta = \frac{N}{3N+1}\left(\frac{P_0}{2KL}\right)^{1/N} N^* \pi$$

The characteristic form can then be derived:

$$\frac{d^2 t}{dV^2} = \frac{d}{dV}\left(\frac{1}{Q}\right) = \frac{d}{dr}\left(\frac{1}{Q}\right)\frac{dr}{dV}$$

$$\frac{d}{dr}\left(\frac{1}{Q}\right) = \frac{d}{dr}\left(\frac{1}{\beta} r^{-(3N+1)/N}\right)$$
$$= -\frac{1}{\beta}\frac{3N+1}{N} r^{-(4N+1)/N} \tag{31}$$

From equation (14), we can find

$$\frac{r}{r_0} = \left(\frac{Q}{Q_0}\right)^{1/4} = \left(1 - \frac{K_s V}{2}\right)^{1/2}$$

which by differentiation yields

$$\frac{dr}{dV} = -\frac{K_s}{4}\frac{r_0{}^2}{r} \tag{32}$$

Therefore, we obtain by multiplication of (31) and (32):

$$\frac{d^2 t}{dV^2} = -\frac{1}{\beta}\frac{3N+1}{N}\left(-\frac{K_s r_0{}^2}{4}\right) r^{-(5N+1)/N} \tag{33}$$

Combining (33) with (30), we find

$$\frac{d^2 t}{dV^2} = \frac{3N+1}{4N} K_s r_0{}^2 \left(\frac{1}{Q}\right)^{(5N+1)/(3N+1)} \beta^{2N/(3N+1)}$$

Noting that $\beta = Q_0/r_0^{(3N+1)/N}$, we finally obtain

$$\frac{d^2 t}{dV^2} = \frac{3N+1}{4N} K_s Q_0^{2N/(3N+1)} \left(\frac{dt}{dV}\right)^{(5N+1)/(3N+1)} \tag{34}$$

which is the characteristic form for a constant pressure standard blocking filtration of a power-law non-Newtonian fluid. Once again, we see that the clogging constant and the exponent are modified by the non-Newtonian property factors

Complete Blocking Law

Combining the flow-rate definition with (28) gives

$$Q = \frac{dV}{dt} = A u_0 = A \frac{N}{3N+1}\left(\frac{P_0}{2KL}\right)^{1/N} r_0^{(N+1)/N}$$

Combining (1) with the above equation yields

$$\frac{d^2 t}{dV^2} = \frac{d}{dV}\left(\frac{1}{Q}\right) = \frac{d}{dV}\left[\frac{1}{(A_0 - \sigma V)u_0}\right] = \sigma u_0 \left(\frac{1}{Q}\right)^2$$

hence

$$\frac{d^2 t}{dV^2} = K_b\left(\frac{dt}{dV}\right)^2$$

with

$$K_b = \sigma u_0 = \sigma \frac{N}{3N+1}\left(\frac{P_0}{2KL}\right)^{1/N} r_0^{(N+1)/N} \tag{35}$$

We see that the exponent is unchanged ($n = 2$) but the clogging constant is dependent upon the non-Newtonian fluid properties.

Intermediate Blocking Law

In this case, we see from equation (9) that the clogging constant K_i is merely a function of the properties of the filter membrane (A_0) and of the solid particles (σ). In no way should K_i be influenced by the type of liquid involved in the filtration process.

Indeed, let us introduce the value of u taken from (28) in the flow rate relation:

$$Q = \frac{dV}{dt} = Au_0 = A\frac{N}{3N+1}\left(\frac{P_0}{2KL}\right)^{1/N} r_0^{(N+1)/N}$$

Let us then differentiate

$$\frac{d^2 t}{dV^2} = \frac{d}{dV}\left(\frac{1}{Q}\right) = \frac{d}{dt}\left(\frac{1}{Au_0}\right)\frac{dt}{dV}$$

Combining this relation with (7) and (9), we obtain

$$\frac{d^2 t}{dV^2} = \frac{d}{dt}\left(\frac{1+K_i Q_0 t}{A_0 u_0}\right)\frac{dt}{dV} = \frac{K_i Q_0}{A_0 u_0}\frac{dt}{dV} = K_i\frac{dt}{dV}$$

Thus, we see that neither the clogging constant K_i nor the exponent ($n = 1$) are modified by the type of fluid. The characteristic form remains exactly the same for Newtonian and non-Newtonian fluids as far as constant pressure filtration is concerned.

SYMBOLS USED

- A filter membrane surface area (m^2)
- C volume of solid particles retained per unit filtrate volume (—)
- d particle equivalent sphere diameter (m)
- K fluid consistency index (non-Newtonian fluids) ($kg\,s^N/m^2$)
- L membrane thickness (m)
- m mass ratio of wet to dry cake (—)
- N flow behaviour index (non-Newtonian fluids) (—)
- N^* number of membrane pores (—)
- P filtration pressure (N/m^2)
- Q flow rate (m^3/s)
- R filter resistance (m^{-1})
- r pore radius (m)
- s mass fraction of solids in slurry (—)
- t filtration time (s)
- u filtrate linear velocity (m/s)
- V filtrate volume (m^3)
- V^* slurry volume (m^3)
- W cake mass (kg)

Greek symbols

- α cake specific resistance (m/kg)
- γ filtrate density (kg/m^3)
- γ_0 solids density (kg/m^3)
- γ_s slurry density (kg/m^3)
- μ filtrate Newtonian dynamic viscosity ($N\,s/m^2$)
- σ blocked area per unit filtrate volume (m^{-1})
- τ shear stress (N/m^2)
- ψ particle shape factor (—)

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ADDRESS

Correspondence about this paper should be sent to Professor J Hermia, Société Belge de Filtration, Secretariat, Institut de Genie Chimique, Université Catholique de Louvain, Voie Minckelers 1, Louvain-la-Neuve, B-1348, Belgique

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